# Observer-Consistent Semantic Quotients for the Ruliad: A Fixed-Cutoff Bridge to Multiway Homotopy and Generalized Observers

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**Abstract**

The ruliad can be thought of as the entangled space of all possible rules and all their consequences. The practical problem is then to figure out which rules can actually yield coherent observer-level physics, and whether that enormous rule space can eventually be narrowed toward a single underlying "God Rule." The difficulty is that the space is vast, while many different microscopic histories and rule families can look similar once seen through finite observers. Here we give a fixed-cutoff bridge from the Wolfram observer/ruliad program to Observer-Patch Holography (OPH). A truncated multiway history is packetized into finite observer data, quotiented by implementation-hiding distinctions, and repaired to a schedule-independent normal form, yielding an explicit observer-sampling functor $S_O^{(L)}$. On this observer side, consistency becomes an equalizer, cycle holonomy becomes an exact rejection test, and benchmark failure becomes a sound negative certificate. The combined picture is then clear: Wolfram Physics supplies the ambient multiway and homotopy structure, while OPH supplies descent, obstruction, and finite search filters that actually narrow rule space. On a four-state toy corpus of 154 rule families at depth $L = 3$, the observer quotient collapses the corpus to 8 classes, and the early filters exclude 62 families, leaving 2 classes on the defect-free target.

## 1 Introduction

The Wolfram-model program has two clear mathematical strengths. First, it supplies a broad multiway and rulial setting in which many rule systems can be studied at once [4, 6]. Second, it has developed a substantial higher-categorical and homotopy-theoretic language for those systems [7, 8, 9]. The remaining weakness is observer semantics at fixed cutoff. Wolfram's observer theory emphasizes equivalencing, boundedness, and persistence [5], while the generalized-observer program of Elshatlawy, Rickles, and Arsiwalla, together with Senchal's extension, formulates minimal observers, observer sampling maps, and a family of boundedness and relevance notions that are explicitly intended to be sharpened further [10, 11]. What is still missing is a theorem-bearing descent semantics that turns those ideas into finite observer tests on rule space.

One can state the issue very directly. A multiway system may branch through many microscopic histories, yet an embedded observer never tracks all of that detail. The observer keeps a bounded local summary, ignores many implementation-level differences, and still expects the resulting meaning to remain stable. So the observer theory has to do three things at once. It has to compress raw histories into observer data. It has to say which microhistories count as the same. And it has to explain why the observer-visible result does not depend on arbitrary update order.

Observer-Patch Holography is useful precisely at that point. OPH starts from local observer algebras, overlap consistency, quotient-defined repair, and record-bearing observer patches [3, 1, 2].

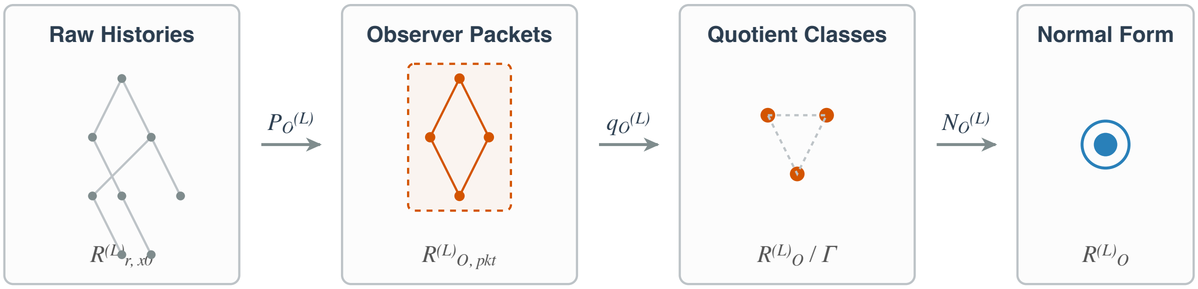


Figure 1: Reader-level view of the fixed-cutoff sampling pipeline. Raw multiway histories are packetized into bounded observer data, quotiented by observer-indistinguishable implementation detail, and repaired to a unique observer-visible normal form.

At fixed cutoff it already contains a schedule-independent normal-form theorem, abelian holonomy obstructions, higher-gauge continuation data, a finite-constraint MaxEnt branch, and an operational observer architecture. Those are the objects needed to turn observer sampling into a concrete functorial pipeline.

This paper develops that bridge. The key composite is

$$R^{(L)}_{r,x_0} \xrightarrow{P_O^{(L)}} R^{(L)}_{O,\mathrm{pkt}} \xrightarrow{q_O^{(L)}} \overline{R}^{(L)}_O \xrightarrow{N_O^{(L)}} R^{(L)}_O,$$

where $P_O^{(L)}$ packetizes truncated multiway histories into finite observer packets, $q_O^{(L)}$ passes to the OPH implementation-hiding quotient, and $N_O^{(L)}$ applies the OPH repair normal form. The resulting functor

$$S_O^{(L)} := N_O^{(L)} \circ q_O^{(L)} \circ P_O^{(L)}$$

is the fixed-cutoff bridge to Senchal's sampling map $S_O : R \to R_O$ itself.

The guiding picture is simple. One starts from microscopic branching histories, records only bounded local observer data, passes from those packets to observer-equivalence classes, and then repairs to the normal form on which physical interpretation is to be read. Much of the paper can be read as showing that these stages can be made mathematically sharp, and that the composite already supports concrete exclusion tests on rule space.

Figure 1 condenses the paper's basic move into one observer-semantic pipeline. The first stage forgets most microscopic detail while keeping bounded local observer data. The second stage identifies histories that differ only by implementation-hiding structure. The third produces the repaired normal form on which physical interpretation is to be read.

### 1.1 Informal roadmap and bridge ingredients

Before turning to the formal definitions, it is useful to say informally what the paper does.

1. Start from the raw Wolfram-model side: all rewrite histories of one rule family $r$ from one initial condition $x_0$ up to one finite cutoff $L$.

2. Read each raw history through a finite observer specification $O$. This produces bounded observer packets. The observer is now working with finite local data on patches and overlaps.

3. Quotient those packets by implementation-hiding data. Microscopic differences that do not matter to the declared observer are identified at this stage.

4. Repair the quotient packet to a unique observer-level normal form. This is where OPH objectivity enters: if repair is schedule-independent, then the observer-visible meaning does not depend on the order in which repair steps are applied.

5. Ask whether the local observer data glue consistently across overlaps. This gives the equalizer formulation of consistency and the cycle-obstruction test for defect-free realizability.

6. Use those observer-visible structures as early rule-search filters. A rule family can then fail by nonunique repair, by nonvanishing holonomy obstruction on an exact tested cycle, or by failure of the finite-constraint branch. This is the operational core of observer-semantic pruning of rule space.

Only a small fixed-cutoff part of OPH is used in this paper. The aim is to isolate the ingredients that matter for the present bridge and to make their role explicit.

1. **(A1) Finite observer cover with overlaps.** Informally, the observer sees reality patchwise, and neighboring patches must agree on shared data. This first enters in Sections 4 and 5, where it enables packet readout, overlap matching, equalizer semantics, and holonomy tests.

2. **(A2) Implementation-hiding quotient.** Informally, observer semantics should ignore microscopic implementation detail. This first enters in Section 4, where raw packet states are replaced by observer-equivalence classes.

3. **(A3) Quotient-defined repair with schedule-independent normal form.** Informally, observer-level content is the repaired quotient normal form, not the unreduced trace. This is the key ingredient in Sections 4 and 5: it makes the sampling functor well-defined and turns objectivity into uniqueness of normal form.

4. **(A4) Finite local constraint family plus record-bearing observer patches.** Informally, stable law and operational observers are finite and local on the realized branch. This enters mainly in Section 6, where it supports the finite-constraint obstruction, the sharpened boundedness/relevance/domain notions, and the operational minimal-observer construction.

5. **Bridge-side locality assumption.** This is not an additional OPH axiom. It is the bridge assumption that one rewrite only forces bounded local packet updates on the observer supports it touches. This is what allows the packetization functor $P_O^{(L)}$ to be defined in Section 4.

So, at the most informal level, the paper uses the bridge ingredients in the following order: local observer patches first, quotienting second, repair-to-normal-form third, overlap-consistency and obstruction theory fourth, and finite observer-side search filters last.

The strongest claims of the paper are all fixed-cutoff claims:

1. an explicit observer-sampling functor $S_O^{(L)}$ with schedule-independent values;

2. an equalizer formulation of OPH consistency together with exact holonomy-based rejection on the defect-free branch;

3. a harness-relative soundness theorem, stated with exact benchmark witnesses and a saturation hypothesis, that turns benchmark failure into a negative certificate for a declared OPH realization class;

4. a finite-constraint obstruction theorem for rule search;

5. concrete OPH realizations of Senchal's boundedness, relevance, domain constraints, interdomain maps, and minimal-observer language.

The paper is strongest in this fixed-cutoff regime. The claims stop short of a canonical packetization for every Wolfram-model substrate, a sufficiency theorem from bounded harness success to recovered-core OPH realization, and an automatic lift from the finite harness to continuum modular geometry or compact gauge reconstruction.

## 2 Assumptions and Claim Boundary

**Assumption 2.1** (Fixed-cutoff OPH basis)**.** *The results below use only the following OPH ingredients:*

*(A1) a finite observer cover $\mathfrak{P} = \{P_i\}_{i\in V}$ with local packet alphabets $S_i$ and overlap alphabets $I_e$;*

*(A2) an implementation-hiding quotient $q : \Sigma_L \to \Sigma_L/\Gamma$ on the finite packet space $\Sigma_L := \prod_{i\in V} S_i$;*

*(A3) a quotient-defined repair law whose normal forms are exactly the consistent quotient packet states, and which is schedule-independent under the OPH termination and local-confluence hypotheses [1];*

*(A4) a fixed finite local constraint family on the realized MaxEnt branch together with record-bearing observer patches [3, 2].*

*Remark* 2.2 (Boundary)*.* The paper keeps three limitations explicit. First, packetization is assumed only for a declared finite observer specification, not for arbitrary substrates. Second, finite benchmark failure yields sound exclusion results, while benchmark success remains only a necessary condition. Third, the continuum and compact-gauge lifts remain conditional exactly where the OPH source papers keep them conditional.

## 3 Notation Concordance with Senchal

To make side-by-side reading possible, this paper now follows Senchal's symbols for the shared observer layer whenever the objects match [11]. The full observer-agnostic ruliad remains $R$. The source used here is the finite slice

$$R^{(L)}_{r,x_0} \subseteq R$$

generated by one rule family $r$, one initial condition $x_0$, and one cutoff $L$.

The bridge factorizes Senchal's sampling functor as

$$S^{(L)}_O = N^{(L)}_O \circ q^{(L)}_O \circ P^{(L)}_O : R^{(L)}_{r,x_0} \to R^{(L)}_O.$$

Here $R^{(L)}_O := S^{(L)}_O(R^{(L)}_{r,x_0})$ is the fixed-cutoff observable ruliad. The three internal OPH maps have no separate names in Senchal's paper because they unpack structure that he leaves implicit: $P^{(L)}_O$ is packetization into bounded observer data, $q^{(L)}_O$ is the implementation-hiding quotient, and $N^{(L)}_O$ is repair to the schedule-independent OPH normal form.

| Senchal notation | OPH bridge meaning in this paper |
|---|---|
| $R$ | The full observer-agnostic ruliad. Same meaning in both papers. |
| $R_O$ | The observer-accessible ruliad. Here this becomes the fixed-cutoff image $R_O^{(L)}$. |
| $S_O : R \to R_O$ | Here realized concretely as $S_O^{(L)} = N_O^{(L)} \circ q_O^{(L)} \circ P_O^{(L)}$. |
| $F_O$ | The actively sampled field inside $R_O$. Here written $F_O^{(L)} \subseteq R_O^{(L)}$. |
| $B(x), P(x), R(x)$ | Boundedness, persistence, and relevance. We use the same symbols below, with OPH formulas. |
| $C_i(x), D_i, S_{i,j}$ | Domain scores, domains, and inter-domain maps. OPH realizes these from explicit finite constraint families and, when needed, separates inclusion from coarse-graining. |

*Remark* 3.1 (Where OPH is finer). Two Senchal symbols compress structure that OPH keeps separate. First, $S_O$ becomes the three-step composite $P_O^{(L)}$, $q_O^{(L)}$, $N_O^{(L)}$. Second, Senchal's inter-domain map $S_{i,j}$ suppresses the distinction between enrichment into a finer presentation and coarse-graining down to a coarser one. OPH keeps those operations distinct because they are implemented by different maps.

# 4 The Observer-Sampling Functor $S_O^{(L)}$

## 4.1 Truncated histories and packet states

Fix a local rule family $r$, an initial condition $x_0$, an observer specification $O$, a finite observer cover $\mathfrak{P} = \{P_i\}_{i \in V}$, and a cutoff $L \in \mathbb{N}$.

**Definition 4.1** (Truncated history category)**.** *The category $R_{r,x_0}^{(L)}$ has as objects the finite rewrite histories*

$$h = (x_0 \xrightarrow{\rho_1} x_1 \xrightarrow{\rho_2} \cdots \xrightarrow{\rho_k} x_k), \qquad 0 \le k \le L.$$

*This is the rule- and cutoff-restricted slice of Senchal's ambient $R$. A morphism $\alpha : h \to h'$ exists iff $h'$ extends $h$ by finitely many rewrites. Composition is concatenation of extension words, and identities are empty extensions.*

In plain language, an object of $R_{r,x_0}^{(L)}$ is just a partial movie of evolution up to depth $L$, and a morphism says that one movie is an initial segment of another. The categorical packaging matters because later we want the observer map to respect extension and composition, not merely assign ad hoc summaries to isolated histories.

**Definition 4.2** (Packet category)**.** *Let $\Sigma_L := \prod_{i \in V} S_i$ be the finite packet space. The category $R_{O,\mathrm{pkt}}^{(L)}$ has packet tuples $s = (s_i)_{i \in V} \in \Sigma_L$ as objects. Its generating morphisms are the declared bounded local update words built from a finite update menu $\mathcal{U}_L$. Morphisms are update words modulo the congruence*

$$\mathbf{u} \sim \mathbf{v} \iff q(\mathbf{u}(s)) = q(\mathbf{v}(s)) \quad \textit{for every } s \in \Sigma_L.$$

*This is an OPH intermediate observer presentation. It sits upstream of Senchal's $R_O$, because quotienting and repair have not yet been applied.*

The packet category is the first place where observer semantics genuinely appears. Instead of storing an entire history, we keep only the patchwise summary the observer is allowed to access. Modding out update words by $\sim$ says that two update procedures count as the same whenever they never lead to distinguishable quotient data. This is the exact mathematical version of the intuitive claim that very different microscopic computations may still look identical to a bounded observer.

For each patch $P_i$, let

$$E_i^{(L)} : \mathrm{Ob}(R_{r,x_0}^{(L)}) \to S_i$$

be the bounded local readout map, and write $E^{(L)} = (E_i^{(L)})_{i \in V}$ for the induced packet readout into $\Sigma_L$.

**Assumption 4.3** (Locality on support)**.** *If $h'$ is obtained from $h$ by one rewrite $\rho$, then only the packets whose supports intersect the rewrite locus need to be refreshed. Equivalently, there is a bounded local packet update word $U(\rho, h)$ such that*

$$U(\rho, h)(E^{(L)}(h)) = E^{(L)}(h').$$

This locality assumption is what keeps the bridge close to the spirit of the Wolfram model. One rewrite is allowed to trigger bounded local observer updates on the supports it touches. Once that is granted, packetization carries a genuine local update structure and not just a brute-force readout of whole histories.

**Definition 4.4** (Packetization functor)**.** *Under the locality-on-support assumption, define*

$$P_O^{(L)} : R_{r,x_0}^{(L)} \to R_{O,\mathrm{pkt}}^{(L)}$$

*by*

$$P_O^{(L)}(h) := E^{(L)}(h)$$

*on objects and*

$$P_O^{(L)}(\alpha_\rho) := [U(\rho, h)]$$

*on one-step extensions, extended to longer morphisms by concatenation.*

**Proposition 4.5** (Functoriality)**.** *The packetization map $P_O^{(L)}$ is a functor.*

*Proof.* The empty extension in $R_{r,x_0}^{(L)}$ is sent to the empty update word, which is the identity in $R_{O,\mathrm{pkt}}^{(L)}$. If

$$h \xrightarrow{\alpha} h' \xrightarrow{\beta} h''$$

are composable history morphisms, then the update taking $h$ to $h''$ is the concatenation of the updates taking $h$ to $h'$ and $h'$ to $h''$. Hence

$$P_O^{(L)}(\beta \circ \alpha) = P_O^{(L)}(\beta) \circ P_O^{(L)}(\alpha).$$

Proposition 4.5 is formally simple and conceptually important. It says that passing from histories to observer packets preserves the extension structure already present on the multiway side. Observer readout can therefore be carried along with evolution itself.

### 4.2 Quotient repair and normal-form semantics

Let $\overline{R}_O^{(L)}$ denote the packet category after passage to the quotient by $\Gamma$, and let

$$q_O^{(L)} : R_{O,\mathrm{pkt}}^{(L)} \to \overline{R}_O^{(L)}$$

be the induced quotient functor.

**Definition 4.6** (Normal-form category)**.** *Let $C_L/\Gamma \subseteq \Sigma_L/\Gamma$ be the quotient consistency set. The category $R_O^{(L)}$ has the quotient normal forms as objects and only identity morphisms. It is the fixed-cutoff observable ruliad in the sense of Senchal.*

A useful way to read $R_O^{(L)}$ is as the observer's dictionary of meaningful states. It is made discrete on purpose: once one has passed to repaired normal forms, the central issue is which observer-visible state remains, not which hidden implementation path was taken to reach it.

By the OPH fixed-point theorem, each quotient packet state has a unique schedule-independent normal form under the termination and local-confluence hypotheses [1]. Let

$$\mathrm{nf}_\lambda : \Sigma_L/\Gamma \to C_L/\Gamma$$

denote this quotient normal-form map.

**Definition 4.7** (Observer-sampling functor)**.** *Define the normal-form functor*

$$N_O^{(L)} : \overline{R}_O^{(L)} \to R_O^{(L)}$$

*by*

$$N_O^{(L)}([s]) := \mathrm{nf}_\lambda([s])$$

*on objects and by sending every morphism to the identity on the target normal form. The observer-sampling functor is*

$$S_O^{(L)} := N_O^{(L)} \circ q_O^{(L)} \circ P_O^{(L)}.$$

**Theorem 4.8** (Schedule-independent observer semantics)**.** *Under Assumption 2.1, item (A3), the functor $S_O^{(L)}$ is well-defined. Its value on every history object is independent of repair schedule.*

*Proof.* The quotient functor $q_O^{(L)}$ is well-defined by construction. The OPH quotient normal-form theorem gives a unique quotient normal form $\mathrm{nf}_\lambda([s])$ for each quotient packet state $[s]$, so the object map of $N_O^{(L)}$ is well-defined. Since $R_O^{(L)}$ is discrete, every morphism is forced to the corresponding identity. Therefore $N_O^{(L)}$, and hence $S_O^{(L)}$, is a well-defined functor. Schedule independence is exactly the OPH uniqueness of the quotient normal form [1].

This is the first main payoff of the bridge. The theorem turns observer objectivity from an intuition into a statement: different asynchronous repair schedules are allowed internally, but they must collapse to the same observer-visible answer.

**Corollary 4.9** (Observer-visible factorization)**.** *Let*

$$M : \mathrm{Ob}(R_O^{(L)}) \to Y$$

*be any observer-visible quantity defined on quotient normal forms. If $h, h' \in \mathrm{Ob}(R_{r,x_0}^{(L)})$ satisfy*

$$S_O^{(L)}(h) = S_O^{(L)}(h'),$$

*then*

$$M(S_O^{(L)}(h)) = M(S_O^{(L)}(h')).$$

*Proof.* The quantity $M$ depends only on the normal-form object. Equal observer semantics therefore imply equal $M$-values.

Corollary 4.9 gives the rigorous version of the rule-search slogan: histories should be compared through observer-visible quotient normal forms, not through their unreduced rewrite traces. In Senchal's notation, this says that observer-level statements should be read on $R_O^{(L)}$, not directly on the raw source slice $R_{r,x_0}^{(L)}$. Once two histories land in the same repaired class, no observer-visible measurement defined on that class can tell them apart.

# 5 Descent, Obstructions, and Sound Negative Certificates

## 5.1 Consistency as a descent object

**Proposition 5.1** (OPH consistency is an equalizer). *Let*

$$C_L = \{s \in \Sigma_L : \pi_{i,e}(s_i) = \pi_{j,e}(s_j)\ \forall e = \{i,j\}\}\,.$$

*Then $C_L$ is the equalizer*

$$C_L = \mathrm{Eq}\left(\prod_{i \in V} S_i \rightrightarrows \prod_{e=\{i,j\}} I_e\right),$$

*where the two arrows are the edgewise restriction maps from the two sides of each overlap.*

*Proof.* An element of $\prod_i S_i$ lies in the equalizer exactly when its restrictions agree on every overlap edge. This is the defining condition for $C_L$.

This equalizer statement can be read in two complementary ways. In plain observer language, one imagines several local patches with shared regions, and consistency means that neighboring patches assign the same overlap data. In categorical language, the same condition says that admissible global packets are exactly the tuples that survive the two competing restriction maps. The equalizer matters because it turns the vague phrase "the local views glue" into a concrete condition that can be checked at fixed cutoff.

This equalizer formulation is the precise point at which OPH enters the higher-categorical Wolfram literature. Arsiwalla and Gorard describe ambient multiway systems in homotopy-theoretic and $(\infty, 1)$-topos language [7, 8]. Proposition 5.1 says what observer-local data must satisfy in order to glue at fixed cutoff, which is the descent condition lurking behind Senchal's passage from $R$ to $R_O$.

## 5.2 Holonomy obstruction

Assume now that each interface alphabet $I_e$ is an abelian group $A$. For a history $h$, define the defect 1-cochain

$$b(h) \in C^1(\mathfrak{P}; A), \qquad b_{ij}(h) := \pi_{i,ij}(E_i^{(L)}(h)) - \pi_{j,ij}(E_j^{(L)}(h)).$$

**Theorem 5.2** (Defect-free holonomy rejection). *If there exists a cycle $C$ in the observer cover such that*

$$\sum_{e \in C} \varepsilon_C(e)\, b_e(h) \neq 0$$

*for some history $h$, then no defect-free global packet assignment exists for $h$. In particular, $h$ is excluded from every defect-free target class.*

*Proof.* The OPH cycle-obstruction theorem states that a global defect-free assignment exists if and only if the signed cycle sum vanishes on every cycle [1]. A nonzero cycle sum therefore obstructs any defect-free global packet assignment.

The intuition is the same as for transporting data around any closed loop. If one walks around a cycle of observer overlaps and comes back with a net discrepancy, then the local descriptions cannot all arise from one globally consistent assignment. The theorem simply states that this intuitive failure can be made exact and computationally usable.

On the genuinely noncentral branch, the same logic upgrades from an abelian cycle sum to a crossed-module class

$$q_\Sigma \in \check{H}^2(N, H \to G),$$

again with vanishing as the criterion for strict global reconciliation [1]. This is the natural fixed-cutoff continuation of the bridge into higher homotopy language.

### 5.3 Harness-relative soundness

**Definition 5.3** (Benchmark harness). *A finite benchmark harness is a tuple*

$$H = (\mathfrak{P}_H, \Sigma_H, \Gamma_H, q_H, \{\pi^H_{i,e}\}, \{\overline{T}^H_i\}, \mathsf{B}_H),$$

*where $\mathfrak{P}_H$ is a finite observer cover, $\Sigma_H$ is the packet space, $q_H : \Sigma_H \to \Sigma_H/\Gamma_H$ is the quotient, $\overline{T}^H_i$ is the benchmark repair interface on quotient packet states, and*

$$\mathsf{B}_H(r) = \mathsf{Confl}_H(r) \wedge \mathsf{Hol}_H(r) \wedge \mathsf{Fin}_H(r)$$

*is the benchmark predicate.*

**Definition 5.4** (Factorization through a harness). *Let $\mathfrak{U}_r$ be a finite-cutoff OPH realization of a rule $r$. The observer-visible neighborhoods of $\mathfrak{U}_r$ factor through $H$ if there is a family of quotient-compatible maps*

$$F^{\mathfrak{U}_r}_{H,N} : N \to \Sigma_H/\Gamma_H$$

*defined on each benchmarked observer-visible neighborhood $N$, preserving the declared overlap projections.*

**Definition 5.5** (Repair-law agreement). *The induced repair law of $\mathfrak{U}_r$ agrees with the benchmark interface on $H$ if, for every benchmarked local update $i$ and every benchmarked neighborhood $N$,*

$$F^{\mathfrak{U}_r}_{H,N} \circ \overline{T}^{\mathfrak{U}_r}_i = \overline{T}^H_i \circ F^{\mathfrak{U}_r}_{H,N}$$

*whenever both sides are defined.*

**Definition 5.6** (Exact benchmark witnesses). *The harness $H$ uses exact benchmark witnesses if the holonomy clause $\mathsf{Hol}_H$ is evaluated on a declared finite family of exact tested cycles and exact overlap-obstruction data. Proxy statistics and surrogate diagnostics are excluded. Equivalently, every failed holonomy witness is a literal nonvanishing obstruction in the sense used by the target OPH branch.*

**Definition 5.7** (Harness saturation)**.** *The harness $H$ is saturated for a declared realization class $\mathcal{C}_H$ if every benchmark failure used by $\mathsf{B}_H$ comes with a declared witness whose benchmark state or tested cycle lies in the image of some map*

$$F^{\mathfrak{U}_r}_{H,N} : N \to \Sigma_H/\Gamma_H$$

*for every realization $\mathfrak{U}_r \in \mathcal{C}_H$ that factors through $H$. In other words, the benchmark is allowed to exclude only on witnesses that are guaranteed to be realized by the factorized observer-visible neighborhoods under consideration.*

**Definition 5.8** (Declared realization class)**.** *Let $\mathcal{C}_H$ be a declared class of finite-cutoff OPH realizations on a chosen branch. The branch specifies which OPH-side necessities are being tested on $H$: schedule-independent quotient repair, a defect-free target branch, and, when relevant, the finite-constraint refinement-stable branch.*

**Theorem 5.9** (Harness-relative soundness)**.** *Let $H$ and $\mathcal{C}_H$ be as above. Suppose a rule $r$ admits an OPH realization $\mathfrak{U}_r \in \mathcal{C}_H$, suppose the observer-visible neighborhoods of $\mathfrak{U}_r$ factor through $H$, suppose the induced repair law agrees with the benchmark interface on $H$, suppose the harness is saturated for $\mathcal{C}_H$, and suppose the holonomy branch of $H$ uses exact benchmark witnesses. Then*

$$\mathsf{B}_H(r) = 1.$$

*Equivalently, if $\mathsf{B}_H(r) = 0$, then $r$ admits no realization in $\mathcal{C}_H$.*

*Proof.* Assume $\mathfrak{U}_r \in \mathcal{C}_H$ but $\mathsf{B}_H(r) = 0$. At least one benchmark clause must fail.

**Confluence failure.** If $\mathsf{Confl}_H(r) = 0$, then the harness supplies a declared benchmark witness consisting of a quotient packet state with two asynchronous repair executions and distinct terminal quotient normal forms. By saturation, that witness lies in the image of some observer-visible neighborhood of $\mathfrak{U}_r$. By repair-law agreement, the benchmark divergence is the image of an induced quotient-repair divergence in $\mathfrak{U}_r$. This contradicts membership in the schedule-independent quotient-repair branch of $\mathcal{C}_H$.

**Holonomy or obstruction failure.** If $\mathsf{Hol}_H(r) = 0$, the harness supplies a declared exact tested cycle whose exact overlap data carry a nonzero obstruction. Because the holonomy witnesses are exact, this is a literal failure of the defect-free criterion on that cycle. By saturation, that exact tested cycle is realized by some factorized observer-visible neighborhood of $\mathfrak{U}_r$. Hence the same exact obstruction is present in the realization, contradicting membership in the defect-free target branch of $\mathcal{C}_H$.

**Finite-constraint failure.** If $\mathsf{Fin}_H(r) = 0$, the harness supplies a declared benchmark witness whose observer-visible image fails the stated finite-constraint criterion. By saturation, that witness is realized by some factorized observer-visible neighborhood of $\mathfrak{U}_r$. Therefore $\mathfrak{U}_r$ cannot lie in the finite-constraint branch carried by $\mathcal{C}_H$.

Every possible benchmark failure contradicts $\mathfrak{U}_r \in \mathcal{C}_H$. Therefore $\mathsf{B}_H(r) = 1$. The contrapositive is immediate.

The theorem is intentionally one-sided. Success on a finite harness is not yet a proof that a rule has the desired physical interpretation. Failure on an exact, saturated harness already rules out membership in the declared realization class. So the harness functions as a disciplined falsification device.

**Corollary 5.10** (Sound negative certificate)**.** *Theorem 5.9 gives exclusion only. Benchmark success on a finite harness is necessary for membership in $\mathcal{C}_H$, not sufficient.*

*Remark* 5.11 (Why saturation and exactness are needed). Factorization through $H$ alone says that realized observer-visible neighborhoods map into the harness. It leaves open whether every benchmark state or every tested cycle used by $\mathsf{B}_H$ is actually realized by those neighborhoods. Saturation supplies that coverage hypothesis. Likewise, the holonomy branch becomes theorem-level only when the tested cycles carry exact overlap-obstruction data. Proxy diagnostics may still be useful, though they play a different role.

# 6 Finite-Constraint Search Filters and Senchal Sharpenings

## 6.1 Finite-constraint obstruction

Fix $L$, and let $\mu_{r,L}$ denote the empirical distribution of packet states in $\Sigma_L$ induced by the histories of $r$ seen by observer $O$. Let $\mathcal{O}_L$ be a chosen finite local observable basis on $\Sigma_L$. Define

$$\kappa_r(L) := \min\left\{|A| : A \subseteq \mathcal{O}_L,\ \exists\lambda \text{ with } \mu_{r,L}(s) = Z^{-1}\exp\Big(-\sum_{O\in A}\lambda_O O(s)\Big)\ \forall s \in \Sigma_L\right\}.$$

**Theorem 6.1** (Finite-constraint obstruction)**.** *If*

$$\sup_{L\geq 1}\kappa_r(L) = \infty,$$

*then $r$ cannot realize the finite-constraint refinement-stable MaxEnt branch of OPH.*

*Proof.* The finite-constraint OPH branch requires one finite local constraint family whose label set does not grow under refinement [3]. Unbounded $\kappa_r(L)$ rules this out.

Theorem 6.1 is a practical rule-search statement. It says that finite observers can stabilize law only on those rule families whose observer-visible image admits a bounded local exponential-family presentation.

The heuristic content is straightforward. If every increase in observational resolution forces one to introduce genuinely new independent constraints, then no finite observer law has stabilized. A candidate rule family may still be interesting, but it is not of the fixed finite-constraint type considered here.

## 6.2 Boundedness, persistence, relevance, and domains

Let $x \in R_O^{(L)}$ be an observer-visible fixed-cutoff state. OPH supplies the following sharpenings of Senchal's vocabulary [11]. The symbols are now the same as his; only the OPH realizations are more explicit. Read operationally, boundedness measures memory cost, persistence measures durability across updates, and relevance asks whether changing a feature can ever alter a later observer-visible readout.

$$\begin{aligned}
&B(x) := \min\{\log_2|S_P| :\ x \text{ is measurable on some accessible patch } P\},\\
&P(x) := \sup\{k \in \mathbb{N} :\ \Pr(X_{t+k} = X_t \mid X_t = x) \geq 1-\eta\},\\
&R(x) = 1 \iff \exists M \in \mathcal{O}_O^{\mathrm{sh}},\ \exists h \text{ such that }\ M(S_O^{(L)}(h \star x)) \neq M(S_O^{(L)}(h)).
\end{aligned}$$

The relevance predicate is especially sharp: relevance is exactly the capacity of a feature to change an observer-accessible quotient-normal-form readout.

*Remark* 6.2 (Threshold convention). Senchal writes boundedness as $B(x) > \beta$. OPH naturally reads $B(x)$ as a resource cost, so the accessible field uses the inequality $B(x) \le \beta$. The symbol is the same; only the monotonic convention differs.

Define the fixed-cutoff field of observation by

$$F_O^{(L)} := \{x \in R_O^{(L)} : B(x) \le \beta,\ P(x) \ge \gamma,\ R(x) = 1\}.$$

This is Senchal's $F_O$, now written on the repaired fixed-cutoff observer image $R_O^{(L)}$.

Let $\mathcal{C} = \{O_a\}_{a=1}^{N_{\text{con}}}$ be the finite OPH local constraint family, and let $\{\mathcal{C}_i\}_{i\in I}$ be any chosen collection of subfamilies indexed by a domain set $I$. When one wants direct notational agreement with Senchal, one may take $I = \{M, S, V, P\}$. For each $i \in I$, define

$$C_i(x) := \min\left\{ |A| : A \subseteq \mathcal{C}_i,\ \begin{array}{l} x \text{ is stable and reconstructible inside the} \\ \text{MaxEnt family generated by } A \end{array} \right\}.$$

Choose a threshold family $(\tau_i)_{i\in I}$ with $\tau_i \ge 0$ for every $i \in I$. If one additionally wants an ordered domain hierarchy, choose a total order on $I$ and require the corresponding thresholds to be monotone along that order. The corresponding OPH domains are

$$D_i := \{x \in R_O^{(L)} : C_i(x) > \tau_i\} \qquad (i \in I).$$

If no domain hierarchy is fixed in advance, the aggregate OPH score is

$$C(x) := \min\left\{ |A| : A \subseteq \mathcal{C},\ \begin{array}{l} x \text{ is stable and reconstructible inside the} \\ \text{MaxEnt family generated by } A \end{array} \right\}.$$

This turns Senchal's domain-threshold language into a concrete OPH observable.

### 6.3 Inter-domain maps $S_{i,j}$

Fix once and for all a finite-dimensional unital $^*$-algebra $\mathcal{A}_*$ equipped with a faithful trace $\tau$. For each constraint family $\mathcal{C}_i$, let $\mathcal{A}_i \subseteq \mathcal{A}_*$ be the finite-dimensional unital $^*$-subalgebra generated by the observables in $\mathcal{C}_i$, and let $\mathsf{D}_i$ be the category whose objects are finite-dimensional $\tau$-observable systems modeled on $\mathcal{A}_i$ and whose morphisms are unital completely positive $\tau$-preserving maps. If $\mathcal{C}_i \subseteq \mathcal{C}_j$, then $\mathcal{A}_i \subseteq \mathcal{A}_j$ is a unital inclusion. In this finite-dimensional traced setting there is a unique $\tau$-preserving conditional expectation

$$E_{j,i} : \mathcal{A}_j \to \mathcal{A}_i,$$

namely the orthogonal projection onto $\mathcal{A}_i$ in the GNS inner product $\langle a, b\rangle_\tau := \tau(a^*b)$.

Senchal writes a single inter-domain map $S_{i,j} : \mathsf{D}_j \to \mathsf{D}_i$. In OPH that symbol splits into two canonical maps, depending on whether one is coarse-graining down or enriching up. In plain language, $J_{i,j}$ is a zoom-in map and $E_{j,i}$ is a controlled forgetting map. The reader can therefore treat $S_{i,j}$ as the coarse-graining branch unless the text explicitly says that enrichment is needed.

Define the enrichment functor

$$J_{i,j} : \mathsf{D}_i \hookrightarrow \mathsf{D}_j$$

by the inclusion $\mathcal{A}_i \subseteq \mathcal{A}_j$ on objects and morphisms, and define the coarse-graining functor

$$E_{j,i} : \mathsf{D}_j \to \mathsf{D}_i$$

by the $\tau$-preserving conditional expectation on objects and by

$$f \longmapsto E_{j,i} \circ f \circ J_{i,j}$$

on morphisms.

For direct comparison with Senchal, one may read

$$S_{i,j} := E_{j,i} \qquad (\mathcal{C}_i \subseteq \mathcal{C}_j)$$

on the downward coarse-graining branch. The upward enrichment branch $J_{i,j}$ has no separate symbol in Senchal's paper, but it is needed in OPH because the inclusion and the conditional expectation are operationally different maps.

**Proposition 6.3** (Functoriality of inter-domain maps)**.** *For nested families $\mathcal{C}_i \subseteq \mathcal{C}_j \subseteq \mathcal{C}_k$,*

$$J_{i,k} = J_{j,k} \circ J_{i,j}, \qquad E_{k,i} = E_{j,i} \circ E_{k,j}, \qquad E_{j,i} \circ J_{i,j} = \mathrm{Id}_{\mathsf{D}_i}.$$

*Proof.* The inclusion identity is tautological. In finite dimension, the $\tau$-preserving conditional expectation is the orthogonal projection in the GNS inner product, and orthogonal projections onto nested subalgebras compose compatibly; this gives $E_{k,i} = E_{j,i} \circ E_{k,j}$. The identity $E_{j,i} \circ J_{i,j} = \mathrm{Id}_{\mathsf{D}_i}$ is the standard retraction property of conditional expectation onto a subalgebra.

## 6.4 Operational minimal observer

The OPH observer patch supplies a concrete tuple

$$O_{\mathrm{OPH}} = (P_O, \mathcal{A}(P_O), \rho_O, \mathrm{Rec}_O, \mathcal{U}_O),$$

where $P_O$ is an observer patch, $\mathcal{A}(P_O)$ its observable algebra, $\rho_O$ its local state, $\mathrm{Rec}_O$ its record layer, and $\mathcal{U}_O$ its local write/compare/update interface [2]. This is the OPH realization of the minimal-observer side of Senchal's formalism, with $\mathrm{Rec}_O$ renamed so it does not collide with Senchal's $R_O$. From this tuple define

$$X_O := \mathrm{Rec}_O, \qquad Y_O := \prod_{e \sim O} I_e, \qquad Z_O := \mathcal{C}_O^{\min},$$

together with the induced state-update map $f_O : X_O \times Y_O \to X_O$ and control map $g_O : X_O \to Z_O$.

**Proposition 6.4** (OPH realizes a generalized minimal observer)**.** *Every fixed-cutoff OPH observer patch determines an operational minimal observer system*

$$\mathsf{Obs}_O^{\mathrm{OPH}} = (X_O, Y_O, Z_O, f_O, g_O, B).$$

*Proof.* The records, overlap packets, correction menus, update maps, and patch boundary data are already present in the OPH screen architecture [2]. Collecting them into the tuple above is immediate.

A reader-level way to parse Proposition 6.4 is this: once the observer has records, incoming overlap data, a finite menu of updates, and a control rule for choosing among them, the generalized minimal-observer language is no longer metaphorical. It is an explicit system extracted from the fixed-cutoff observer patch itself.

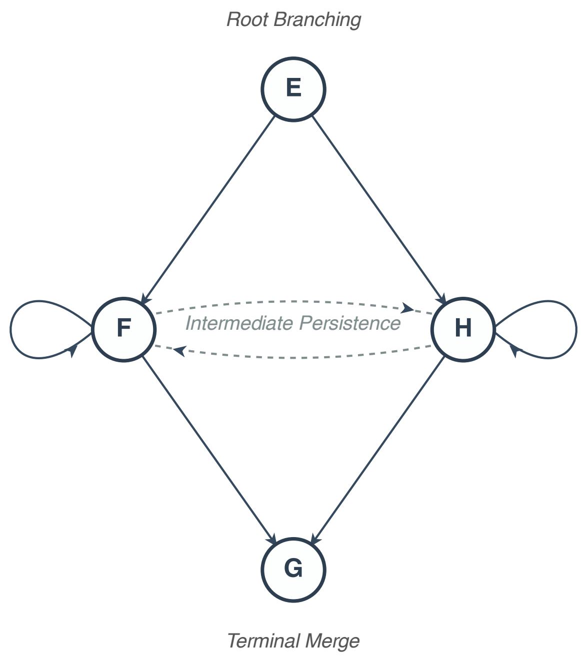


Figure 2: Microscopic motif underlying the toy corpus. The root state branches from $E$ to $F$ or $H$, the intermediate states may persist or transition across, and both branches can merge again at $G$.

# 7 A Fixed-Cutoff Toy Harness

This section records a pilot benchmark whose purpose is modest: to show that the bridge can function as a concrete observer-semantic pruning mechanism on a finite observer-visible corpus.

## 7.1 Corpus and harness

Take the four-state toy space

$$V = \{E, F, H, G\},$$

with initial state $E$, depth cutoff $L = 3$, and the 8 directed elementary rules

$$E \to F, \quad E \to H, \quad F \to G, \quad H \to G, \quad F \to F, \quad H \to H, \quad F \to H, \quad H \to F.$$

Let $\mathcal{R}^{\mathrm{toy}}_{3,4}$ be the collection of all reachable rule families of sizes 2, 3, or 4 drawn from those edges. The audited benchmark notes report

$$|\mathcal{R}^{\mathrm{toy}}_{3,4}| = 154.$$

This toy model is small enough to see in one glance. The key microscopic feature is a familiar Wolfram-style diamond: branching away from $E$, possible persistence on the intermediate states, and eventual merging at $G$. The observer-side question is whether that visible branching can be reconciled consistently across a finite cover.

Figure 2 isolates the small piece of multiway structure that matters for the toy benchmark. It makes the bridge problem concrete: a visibly branching microscopic motif is being tested for whether its observer-side summaries can still be glued into one coherent packet.

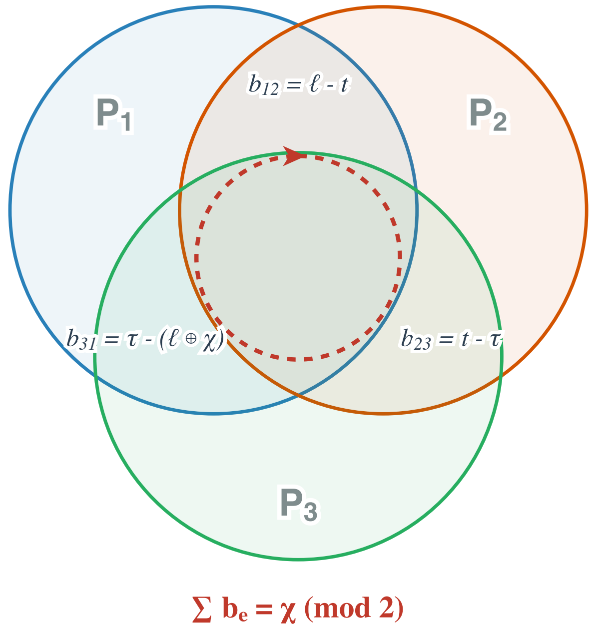


Figure 3: Observer-side cover for the toy holonomy witness. The three patch overlaps carry defects $b_{12}$, $b_{23}$, and $b_{31}$, and the cycle sum recovers $\chi$ mod 2. When both root branches are present, $\chi = 1$, the overlap data fail the defect-free test.

Use a three-patch harness with packet tuple

$$((\ell, \chi), t, \tau) \in (\mathbb{Z}_2 \times \mathbb{Z}_2) \times \mathbb{Z}_2 \times \mathbb{Z}_2,$$

where $\ell$ records whether the first step is $E \to H$, $\chi$ records whether both root branches $E \to F$ and $E \to H$ are present, $t$ records whether $G$ is reached by depth $L$, and $\tau = t$ duplicates the terminal packet. Define the overlap defects in $A = \mathbb{Z}_2$ by

$$b_{12} = \ell - t, \qquad b_{23} = t - \tau, \qquad b_{31} = \tau - (\ell \oplus \chi).$$

Then the oriented cycle sum is

$$b_{12} + b_{23} + b_{31} = \chi \pmod 2.$$

So branching at the root is an exact defect-free holonomy witness on this toy harness.

Figure 3 gives the observer-side half of the same argument. A small microscopic branching-and-merging motif is translated into finite observer packets, and a seemingly qualitative statement about “incompatible views” becomes an exact cycle obstruction.

## 7.2 Measured outcome

Two rule families are declared semantically equivalent when they induce the same reachable packet support and the same set of reachable normal forms. The audited benchmark notes report that the 154 raw rule families collapse to 8 observer-visible semantic classes under this equivalence.

The same notes record the following early-filter counts:

| stage | survivors | rejected at stage | cumulative rejection |
|---|---|---|---|
| raw families | 154 | – | – |
| after confluence filter | 103 | 51 | 33.1% |
| after holonomy filter | 92 | 11 | 40.3% |
| after toy finite-constraint filter | 92 | 0 | 40.3% |

Equivalently,

$$\mathrm{RR}^{\mathrm{toy}} = 62/154 \approx 40.3\%.$$

At the semantic-class level, 6 of the 8 classes are excluded, leaving 2 classes on the defect-free target.

*Remark* 7.1 (Interpretation)*.* These figures should be read as a pilot fixed-cutoff measurement, not as an estimate for realistic Wolfram-model corpora. Their value lies elsewhere: the observer quotient is large even on a tiny corpus, the early observer-side filters are computable from finite observer-visible data, and the combination already removes a substantial fraction of raw families before any continuum or gauge-lift stage is attempted. In that sense, the toy harness is a concrete example of observer-semantic pruning of rule space.

## 8 Relation to Current Wolfram Physics

The cleanest division of labor is now visible. Arsiwalla and Gorard provide the ambient higher structure: multiway systems, higher homotopies, $n$-fold categories, $\infty$-groupoids, and the suggestion of an $(\infty, 1)$-topos-like classifying space [7, 8]. OPH adds the observer-side descent semantics: finite covers, equalizers, obstruction classes, and quotient normal forms. In that sense the ambient space comes from Wolfram Physics, while the gluing criterion comes from OPH.

Wolfram's observer theory is centered on equivalencing, boundedness, and persistence [5]. The bridge developed here turns that language into a concrete finite-cutoff composite. Equivalencing becomes quotient-level repair to a schedule-independent normal form, boundedness becomes finite packet readout on declared patches, and persistence becomes stable record-bearing observer dynamics. Senchal's sampling map is then no longer only a slogan; it is implemented by

$$S_O^{(L)} = N_O^{(L)} \circ q_O^{(L)} \circ P_O^{(L)}$$

[11].

The rule-search gain is substantial. The Wolfram literature already supplies sophisticated global mathematics for multiway evolution [6, 9]. OPH contributes finite observer-side rejection criteria: holonomy obstructions, harness-relative negative certificates, and finite-constraint filters. These are early filters. They do not solve the whole search problem. But they do turn observer semantics into something operational. One can use it to prune rule space in a disciplined way.

If this perspective scales beyond toy harnesses, the gain is not merely rhetorical. It would give the ruliad program a concrete middle layer between raw multiway possibility and candidate physically coherent rule families: a layer where one can test whether finite observers can actually glue, stabilize, and agree before attempting harder continuum reconstructions.

# 9 Conclusion

OPH provides a fixed-cutoff observer semantics for the Wolfram ruliad program. It does so by packetizing multiway histories, quotienting implementation-hiding data, repairing to a unique OPH normal form, and reading physical content only at that quotient-normal-form level.

This yields several concrete gains. The bridge gives a functorial realization of observer sampling, a descent formulation of consistency, exact obstruction tests on the defect-free branch, a sound negative certificate relative to a declared harness, and explicit OPH sharpenings of Senchal's observer vocabulary. It also suggests a more useful style of observer theory for the Wolfram program. One wants an observer theory that says exactly what is kept, what is forgotten, and where gluing can fail. The continuum and compact-gauge branches remain conditional. The fixed-cutoff bridge already stands on its own, and that is enough to make this line of work a potentially valuable component of Wolfram Physics rule search.

# A Future Bridge Theorems and Concrete Next Steps

The main body of the paper stays narrow: finite cutoff, explicit observer packets, exact overlap tests, and sound exclusion criteria. The long-range ambition is stronger. One would like a mathematically controlled way of narrowing the ruliad's rule space: first to exact observer-semantic survivors, then perhaps to a single universal generative rule, or at least to a terminal observer-semantic object that deserves, informally, the name of a "god-rule." This appendix records a menu of bridge statements that seem worth pursuing next. None is claimed as proved here. The purpose is to mark out concrete mathematical improvements and to make the route from finite observer filters to a much sharper narrowing of rule space more definite.

## A.1 Harness survivor quotients

The first improvement is to replace the ambient observer space by a finite exact survivor quotient tailored to one benchmark family. In practice this is the first serious narrowing step on the way from the full ruliad rule space to a much smaller family of observer-semantic survivors.

**Definition A.1** (Harness survivor quotient)**.** *Let $H$ be a fixed exact harness as in Section 5. Define $\mathcal{S}_H$ to be the finite set of semantic equivalence classes of harness-realizable observer states that survive exact benchmark filtration on $H$. Assume:*

1. *exact benchmark witnesses are used;*
2. *the harness is saturated for the benchmark family under discussion;*
3. *survivor classes are identified only up to the declared semantic quotient.*

This is the cleanest finite observer-side replacement for Senchal's abstract $R_O$ when one wants theorem-bearing convergence statements [11]. It is finite, observer-visible, benchmarked, and externally checkable. It also fits the present paper well, because the whole bridge is built around finite packet spaces, quotient descent, and exact exclusion witnesses.

## A.2 Deterministic convergence candidates

**Candidate Theorem A (Finite-time convergence on survivor classes).** Let

$$ER_H : \mathcal{S}_H \to \mathcal{S}_H$$

be a deterministic refinement operator. Suppose there exists a function

$$L : \mathcal{S}_H \to \mathbb{N}^k$$

with lexicographic order such that, for every non-fixed class $[x]$,

$$L(ER_H([x])) <_{\text{lex}} L([x]).$$

Then every orbit of $ER_H$ reaches $\text{Fix}(ER_H)$ in at most $|\mathcal{S}_H|$ steps.

This is the most direct bridge from the current paper to a genuine convergence theory. The present paper already gives the ingredients from which a Lyapunov functional could be built: exact failed-test counts, overlap defects, semantic ambiguities after quotienting, and coarse complexity measures. The finite survivor quotient then supplies the finiteness hypothesis that deterministic convergence arguments actually need.

The most natural candidate is a defect vector of the form

$$L([x]) = \big(N_{\text{fail}}([x]), H_{\text{cyc}}([x]), N_{\text{amb}}([x]), C_{\text{desc}}([x])\big).$$

Read informally, one first tries to remove exact local failures, then nonlocal cycle obstruction, then residual semantic nonuniqueness, and only after that minimize descriptive overhead.

**Candidate Theorem B (Defect descent to exact survivors).** Let

$$D([x]) := \big(N_{\text{fail}}([x]), H_{\text{cyc}}([x])\big).$$

Suppose $ER_H$ is a local refinement rule on $\mathcal{S}_H$ such that:

1. $ER_H$ preserves harness realizability;
2. $ER_H$ never increases $D$;
3. every non-fixed class admits at least one refinement with strict decrease in $D$.

Then every orbit converges in finite time to the set

$$\mathcal{Z}_H := \{[x] \in \mathcal{S}_H : D([x]) \text{ is minimal}\}.$$

This statement is weaker than uniqueness, and it is still powerful. It says the dynamics lands in the defect-minimal exact-survivor sector. That would already be a real synthesis result. The bridge developed in the main text supplies the admissible observer-semantic state space and the exact defect carriers. A Senchal-style refinement dynamics would then supply the direction of travel.

### A.3 Stochastic convergence candidates

**Definition A.2** (Admissible noisy kernel)**.** *Let $K_H$ be a Markov kernel on $\mathcal{S}_H$. Call $K_H$ admissible if it:*

1. *has support only on harness-realizable classes;*
2. *strongly favors defect-lowering moves;*
3. *allows only rare lateral moves between near-equivalent survivors;*
4. *is primitive on the defect-minimal sector $\mathcal{Z}_H$.*

**Candidate Theorem C (Unique stationary survivor law).** If $K_H$ is admissible and primitive on $\mathcal{Z}_H$, then there exists a unique stationary distribution

$$\mu_H^* \in \Delta(\mathcal{Z}_H)$$

such that, for every initial distribution supported on $\mathcal{S}_H$,

$$d_{\mathrm{TV}}(\mu_0 K_H^n, \mu_H^*) \to 0$$

exponentially.

This is the most realistic route to uniqueness. A primitive Markov chain on a small exact survivor sector is much more plausible than a primitive chain on the whole observer-accessible ruliad. It also matches the usual scientific situation more closely. Deterministic refinement may leave several clean survivor classes. Weak noise or exploration can still induce a unique stationary law over them.

### A.4 Terminal survivor objects and the universal-rule question

**Definition A.3** (Terminal survivor object)**.** *Let $\mathcal{Z}_H$ be the defect-minimal survivor sector. A class $T_H \in \mathcal{Z}_H$ is a terminal survivor object if, for every $[x] \in \mathcal{Z}_H$, there exists a unique admissible semantic-refinement morphism*

$$[x] \to T_H$$

*inside the survivor category.*

**Candidate Theorem D (Terminality inside the survivor category).** Suppose $\mathcal{Z}_H$ admits a terminal survivor object $T_H$. Then every deterministic fixed point and every stochastic equilibrium supported on $\mathcal{Z}_H$ maps uniquely to $T_H$.

This is the right formal place to talk about a "god-rule" direction without overclaiming. The current paper does not prove any unique terminal object. If a terminal survivor object can be exhibited inside a benchmarked observer-semantic category, one has a much more meaningful universality statement than an ambient terminality claim on the whole ruliad. The formal object would still be observer-relative and finite-cutoff. It would now sit at the end of a real filtration-and-refinement pipeline.

### A.5 Ratchets, quotient descent, and law selection

**Candidate Theorem E (Semantic ratchet on harness survivors).** Let $PS_H(t) \subseteq \mathcal{S}_H$ be the set of exact defect-minimal survivor classes discovered by time $t$. Suppose:

1. survivor classes, once discovered, remain admissible under subsequent exact harness checks;
2. $ER_H$ preserves exact survivorhood;
3. the harness coverage assumptions remain fixed.

Then, for $t_1 \le t_2$,

$$PS_H(t_1) \subseteq PS_H(t_2).$$

This would turn Senchal's ratchet idea into something much sharper. It would become a theorem candidate about exact benchmark-admissible semantic survivors.

**Candidate Theorem F (Quotient compatibility theorem).** Let

$$q : \Sigma_H \to \mathcal{S}_H$$

be a semantic quotient from raw harness states to survivor classes, let

$$\mathrm{nf}_H : \Sigma_H \to \Sigma_H$$

be a repair or normal-form map, and let $ER_H$ be a class-level update rule. Assume

$$q \circ \mathrm{nf}_H = ER_H \circ q.$$

If $\mathrm{nf}_H$ is terminating and locally confluent on the admissible harness region, then class-level observer dynamics are schedule-independent and descend to a unique survivor normal form on $\mathcal{S}_H$.

This is especially attractive from the point of view of the present paper, because it would tie any future class-level convergence theorem directly back to quotient repair on raw packet states. Then observer dynamics would converge as the quotient image of an underlying repair dynamics with a genuine normal-form theorem.

**Candidate Theorem G (Two-class endpoint sector).** Suppose the exact filtered survivor quotient $\mathcal{S}_H$ has exactly two defect-minimal classes. Then the strongest safe claims are the following:

1. deterministic survivor dynamics may converge to either class, depending on basin;
2. admissible primitive stochastic dynamics yield a unique stationary distribution supported on those two classes;
3. uniqueness of a single terminal survivor object requires an additional theorem showing either:
   (a) one class strictly refines the other,
   (b) both map uniquely to a common terminal survivor object, or
   (c) both are presentations of one deeper semantic object.

This point matters for scientific support. A toy harness with two clean defect-free classes is already a useful achievement. It still falls short of a proof of unique universal selection.

**Candidate Theorem H (Selection on survivor-law space).** Let $\Lambda_H$ be the finite set of harness-admissible survivor-update laws. Define a fitness

$$f(\lambda) = -\alpha\, N_{\mathrm{fail}}(\lambda) - \beta\, H_{\mathrm{cyc}}(\lambda) + \gamma\, R_{\mathrm{rob}}(\lambda) - \delta\, K(\lambda).$$

Under replicator dynamics on $\Lambda_H$, mean fitness is nondecreasing. Hence exact, holonomy-clean, robust, low-complexity survivor laws are dynamically selected.

This would shift attention from single states to update laws themselves. That may ultimately be closer to the real universal-rule problem. One would then ask which law templates continue to survive refinement while remaining exact, stable, and simple.

### A.6 Most concrete next steps

If one wants the strongest realistic bridge program, the right order is:

1. define the finite survivor quotient $\mathcal{S}_H$ for a concrete benchmark family;
2. define an explicit defect functional $D$ from exact test failures, holonomy data, and semantic nonuniqueness;
3. define the admissible move set $ER_H$ and prove deterministic finite-time convergence to the defect-minimal sector;
4. add admissible noise and prove uniqueness of a stationary law on that minimal sector;
5. only then ask whether terminality emerges inside the survivor category.

That ordering matters. It keeps the mathematics in the same disciplined progression as the present paper itself: finite observer state space first, exact filtration second, convergence third, stochastic uniqueness fourth, and universal interpretation only at the end. If this program succeeds, the result would not yet be a proof that a single rule literally generates everything in one stroke. It would, however, be a serious path toward an observer-semantic theory of why the space of viable rules might collapse toward one terminal generative object.

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